

OPTIMAL ROUTE IN JAKARTA

FINDING SYSTEM

ALGORITHMS AND DATA
STRUCTURES



MEET OUR TEAM

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ALGORITHM

1. Graph-Based Approach

We use a weighted graph to represent the map and roads:

- Nodes (Cities) → each location such as Monas, Kota Tua, Senayan, etc.
- Edges (Roads) → each road has distance and traffic level.
- Edge Weight Formula:
 - $\text{finalWeight} = \text{distance} \times (1 + \text{trafficLevel})$
 - This ensures that congested roads become more "expensive" to travel through.

2. Dijkstra's Algorithm

Purpose in Our System

Used when the system needs:

- The absolute shortest path
- Without using any heuristic
- Purely based on the total accumulated weight

How Dijkstra Works

- Starts from the source node
- Continuously chooses the node with the smallest known distance
- Expands all neighbors and updates distances
- Guaranteed to find the true optimal path
- Best for uniform exploration or when the heuristic is not useful



ALGORITHM



3. A* Search Algorithm

How A* Works

- Uses the combined cost:
 - $g(n)$ → actual cost from start
 - $h(n)$ → heuristic distance estimate
 - $f(n) = g(n) + h(n)$
- The node with the smallest $f(n)$ is expanded
- Reaches the goal faster because it searches toward the destination

Dijkstra

- Finds the true shortest path
- No heuristic
- Thorough but slower
- Ideal for accuracy

*A

- Uses distance + heuristic
- Much faster in real maps
- Ideal for navigation
- Chooses the path with lowest $f(n) = g + h$



THANK
YOU

